Square-Wave Voltammetry Forward Model

How MMP-9 Concentration Becomes a Measured Voltammogram

Eton iGEM 2026, Dry Lab | Layers 8, 10, 11, 13, 15 | `swv_forward_model.py`

Scope. This document presents the forward model describing how MMP-9 activity at the electrode surface is transduced into a measured voltammogram. It is organised into six sections, each corresponding to one physical step in the signal chain from probe to peak. Each section follows a consistent structure: a plain-language description of the process; the assumptions required to obtain the stated result, given explicitly; the governing equation, with its physical interpretation stated before its derivation; the relevant parameters; and the resulting prediction. Values obtained from a published source are annotated *(confirmed)*. Values that represent a working assumption not yet confirmed on the fabricated device are annotated *(unconfirmed)*.

I. Overview: Electrode Structure and Operating Principle

The sensor is a gold electrode with a geometric area of 1.7 mm^2 (0.017 cm^2) [12] *(confirmed)*, which supersedes an earlier 1.5 mm-diameter design specification (0.0177 cm^2); the ∼4% discrepancy is too small to alter any qualitative conclusion in this report. A single probe type is attached to the electrode surface via a thiol (sulfur) anchor (Figure 1): a ten-residue peptide (SGKIPRTLTA) that constitutes the MMP-9 cleavage site, connected to the gold via a PEG6 spacer, and terminating in a covalently conjugated methylene blue (MB) label. MB undergoes reversible electron transfer with the gold surface, which constitutes the measured signal. Upon cleavage of the peptide by MMP-9, the MB-bearing fragment is released into solution and the resulting truncated probe is no longer electroactive. An earlier configuration of this device additionally incorporated a ferrocene (Fc) reference probe alongside MB for a ratiometric correction; Fc has since been removed from the protocol [12], for reasons detailed in Part E.

As MMP-9 activity increases, the quantity of intact MB-bearing probe decreases, and the corresponding voltammetric peak diminishes in proportion. The extent of this decrease, measured on a single electrode before and after a defined incubation period, constitutes the readout of enzymatic activity from which [MMP-9] is subsequently estimated.

The remainder of this document is organised into six sections, each addressing one stage of this signal chain:

Part	Content	Section
A	Composition of the electrode surface	II
B	Spatial distribution of the MB label	III
C	Origin of the signal from a single MB label	IV
D	Justification for square-wave voltammetry	V
E	Derivation of the cleavage fraction from the measured peak	VI
F	Relationship between [MMP-9] and the predicted signal	VII

Figure 0: Physical electrode assembly and tether geometry

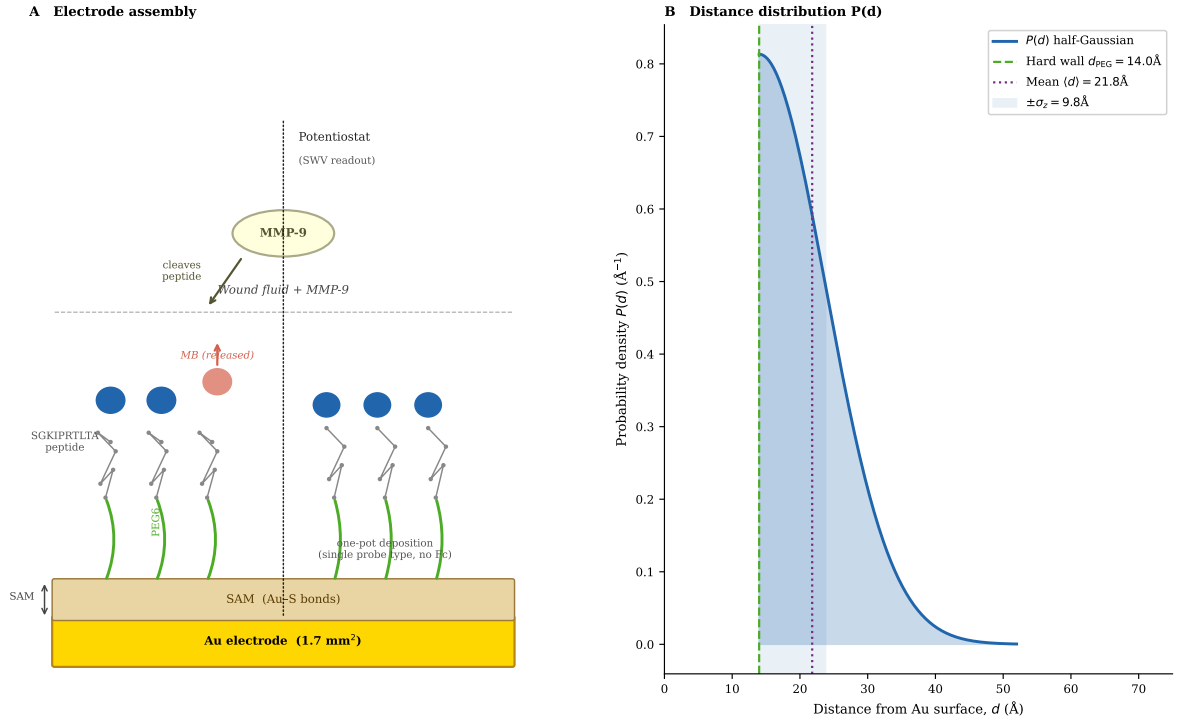


Figure 1: **Left:** the MB-peptide probe on the gold surface (single probe type; no Fc reference, Part VI). **Right:** the range of distances the MB label can sit at above the gold, set by the flexibility of its tether (Part III).

II. Part A: The Surface Chemistry

Overview

An accurate description of the electrode surface composition is a prerequisite for the electrochemical treatment that follows. This section supersedes an earlier, incorrect description. Previous versions of this report, and related layer reports, described a 6-mercapto-1-hexanol (MCH) backfill step intended to fill unoccupied sites between probes, together with a two-component deposition incorporating a ferrocene (Fc) reference probe. Neither description reflects the current protocol. The protocol in current use [12] deposits a single probe species, MB-peptide only, with no MCH and no Fc.

Assumptions

- **Single-component deposition (no MCH, no Fc).** This is a confirmed property of the fabrication protocol [12], not a modelling choice.
- **Total probe coverage $\Gamma_{\text{tot}} = 3 \times 10^{-12}$ mol cm^{-2}.** This value has not been measured for the electrodes in use; it is an engineering estimate within the range reported for a comparable peptide-PEG system in the literature [8] *(unconfirmed)*. Direct measurement is not possible with the instrumentation currently available to the team [12], so this value is expected to remain a placeholder for the foreseeable future. Any prediction sensitive to the exact value of Γ_{tot} should be interpreted accordingly.
- **Exposure of bare gold between probes.** In the absence of an MCH backfill, regions of bare gold are likely to remain exposed between probes, presenting a plausible route for non-specific protein adsorption from wound fluid (fouling). This has not been experimentally assessed *(unconfirmed)*.

The distance distribution $P(d)$ used in Part III was derived under the assumption of an isolated, PEG6-spaced probe with no neighbouring species. This assumption has not been re-derived or validated for the present MCH-free surface, and $P(d)$ should accordingly be regarded as a plausible working form rather than a measured quantity.

III. Part B: Position of the MB Label Above the Electrode

Overview

The MB label is not rigidly fixed above the gold surface: it is positioned at the terminus of a flexible tether (the PEG6 spacer together with the peptide itself) and consequently samples a distribution of distances. The rate of

electron transfer decreases approximately exponentially with distance, consistent with standard Marcus/tunnelling theory [9,10], such that only probes positioned close to the gold contribute appreciably to the measured signal.

Assumptions

- **The tether behaves as a freely-jointed chain (FJC).** This is a standard polymer-physics model for a flexible, non-rigid linker, used here for the peptide segment of the tether. It is a modelling choice, not a measurement on our specific peptide.
- **This geometry has not been validated on the real (MCH-free) surface.** See Part II: the calculation below assumes a clean, isolated PEG6-spaced probe, which has not been directly confirmed on the fabricated electrode *(unconfirmed)*.

Derivation

Chain statistics. The peptide segment of the tether (SGKIPRTLTA, $N_{\text{res}} = 10$ residues) is treated as a freely-jointed chain (FJC): a sequence of N_K independent, freely-orienting rigid (Kuhn) segments, the standard coarse-grained model for a flexible, unfolded polymer [3]. For a polypeptide backbone, the contour length per residue is $l_0 \approx 0.38$ nm and the Kuhn length is $b \approx 0.76$ nm (two residues per Kuhn segment) [3], giving

$$N_K = \frac{N_{\text{res}}\, l_0}{b} = \frac{10 \times 0.38}{0.76} = 5 \text{ Kuhn segments.} \tag{III.1}$$

Because the N_K segments are oriented independently and isotropically, the end-to-end vector $\mathbf{R}$ of the chain is a sum of N_K independent, identically distributed random vectors. By the central limit theorem, for $N_K \gg 1$ each Cartesian component of $\mathbf{R}$ is approximately normally distributed with mean zero and variance

$$\langle R^2 \rangle = N_K b^2, \qquad \sigma_i^2 = \frac{\langle R^2 \rangle}{3} \quad (i = x, y, z), \tag{III.2}$$

the factor of 3 arising because the total mean-square displacement is shared equally among the three Cartesian directions by isotropy. Numerically,

$$\langle R^2 \rangle = N_K b^2 = 5 \times (0.76 \text{ nm})^2 = 2.888 \text{ nm}^2, \qquad \sigma_z = \sqrt{\langle R^2 \rangle / 3} = 0.981 \text{ nm},$$

where z is the coordinate normal to the gold surface, along which distance from the electrode is measured.

Reference point. The peptide is not anchored directly to the gold; it is attached via a PEG6 spacer, treated as a second, independent FJC segment of 6 ethylene-oxide units ($l_{\text{EO}} = 0.35$ nm per unit, same Kuhn length $b = 0.76$ nm), giving $n_{K,\text{PEG}} = 6 \times 0.35/0.76 = 2.76$ Kuhn segments. Because this spacer is short, its equilibrium extension is estimated using Flory's excluded-volume scaling law for a self-avoiding chain in a good solvent,

$$R_F = n_K^{3/5}\, b, \tag{III.3}$$

rather than the ideal-chain ($R \propto N^{1/2}$) result used for the longer, more nearly ideal peptide above. This gives $R_F = 2.76^{3/5} \times 0.76$ nm $= 1.40$ nm, taken as the mean distance d_{PEG} of the PEG6 tip from the gold surface.

Resulting distribution. The MB label sits at the tip of the peptide segment, itself rooted at the PEG6 tip at mean distance d_{PEG} from the gold. Since the gold surface is impenetrable, only configurations with $d \geq d_{\text{PEG}}$ are physically accessible; folding the Gaussian distribution of the z-displacement (variance σ_z^2, Eq. (III.2)) onto this half-line and renormalising gives the half-Gaussian distance distribution

$$P(d) = \sqrt{\frac{2}{\pi}}\, \frac{1}{\sigma_z} \exp\!\left(-\frac{(d - d_{\text{PEG}})^2}{2\sigma_z^2} \right), \qquad d \geq d_{\text{PEG}}, \tag{III.4}$$

with $d_{\text{PEG}} = 1.40$ nm and $\sigma_z = 0.981$ nm as derived above. The prefactor $\sqrt{2/\pi}$ restores $\int_{d_{\text{PEG}}}^{\infty} P(d)\, \mathrm{d}d = 1$, since an unrestricted Gaussian of standard deviation σ_z has exactly half its area on each side of its mean.

Electron-transfer rate versus distance. MB undergoes non-adiabatic electron transfer with the gold electrode by quantum-mechanical tunnelling through the intervening tether. Marcus theory predicts that the electronic coupling element, and hence the standard rate constant k^0, decays exponentially with tunnelling distance [9,10]:

$$k^0(d) = k^0(0)\, e^{-\beta d}, \tag{III.5}$$

where β is the tunnelling decay constant of the intervening medium and $k^0(0)$ is the (extrapolated, not physically attainable) rate at zero distance. For alkanethiol-tethered ferrocene and methylene blue, $\beta = 1.0$ Å$^{-1} = 10$ nm^{-1} is a confirmed literature value [4]; no PEG-specific value for β exists in the papers consulted for this report, so the alkanethiol value is used as a placeholder *(unconfirmed)*. Dauphin-Ducharme (2017) [1] report $k^0(0) = 1.5 \times 10^8$ s^{-1}

for MB directly at the electrode (their Figure 4 extrapolation). Evaluating Eq. (III.5) at the PEG6 tip, $d = d_{\mathrm{PEG}} = 1.40$ nm, gives the reference rate used throughout this report:

$$k^0_{\mathrm{ref}} = k^0(d_{\mathrm{PEG}}) = 1.5 \times 10^8 \times e^{-10 \times 1.40} = 127\ \mathrm{s}^{-1} \quad \textit{(unconfirmed)}. \tag{III.6}$$

Because $k^0(d)$ falls by roughly an order of magnitude for every ∼2.3 Å of additional distance, MB labels only slightly further from the PEG6 tip than average contribute negligibly to the measured signal, even though such distances are not rare under $P(d)$; this is the origin of the "near probes dominate" behaviour shown in Figure 2B.

Operating point

At the confirmed operating frequency of 10–20 Hz [12], Eq. (III.6) corresponds to $\kappa_{\mathrm{tip}} = k^0_{\mathrm{ref}}/(2f) = 6.4$ at 10 Hz or 3.2 at 20 Hz, both within the quasi-reversible regime (Part IV) in which the signal is largest, although the precise value depends on the frequency employed for a given scan. An earlier version of this report assumed 15 Hz, giving $\kappa_{\mathrm{tip}} = 4.2$; this value has been superseded by the confirmed operating frequency.

Result

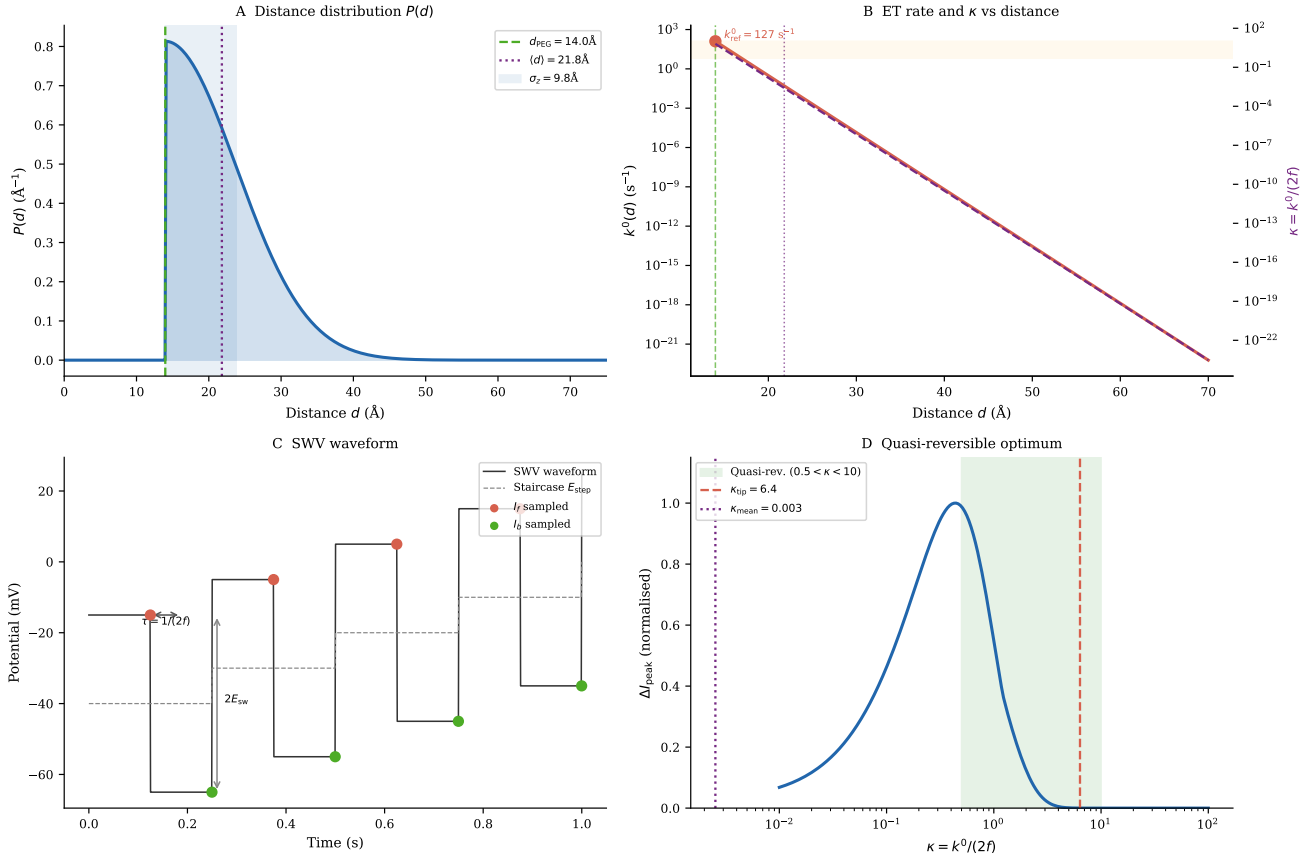


Figure 2: **A:** the distance distribution $P(d)$; most MB labels sit within about 1 nm of the PEG6 spacer tip. **B:** the electron-transfer rate $k^0(d)$ falls off rapidly with distance, so nearby probes dominate the signal. **C:** the square-wave voltammetry (SWV) pulse pattern used to scan the electrode. **D:** the net signal peaks at an intermediate scan speed: too fast, and probes cannot react in time; too slow, and forward and backward pulses give the same answer and cancel.

IV. Part C: The Core Signal Model

Overview

Each MB label transitions between oxidised and reduced states at a rate determined by the applied potential, following Butler-Volmer kinetics, the standard model for electrode reactions [9]. Because MB is confined to the electrode surface rather than diffusing freely in solution, the governing equation for the fraction of MB in the reduced state admits a single, exact closed-form solution, obtained without recourse to numerical simulation.

Assumptions

- **Butler-Volmer kinetics apply to surface-confined MB.** This is the standard, textbook model for a redox species attached to an electrode, not specific to our device, and is not in question here.

- **The potential is held fixed within each SWV half-pulse.** This is true by construction of the SWV waveform (Part V) and is what makes the equation below exactly solvable.

Equations

The rate of change of reduced-MB coverage Γ_{red} follows:

$$\frac{d\Gamma_{\mathrm{red}}}{dt} = k_f\,(\Gamma_{\mathrm{tot}} - \Gamma_{\mathrm{red}}) - k_b\,\Gamma_{\mathrm{red}} \tag{IV.1}$$

Because k_f, k_b are constant during a half-pulse, this has a closed-form solution:

$$\boxed{\Gamma_{\mathrm{red}}(\tau) = \Gamma_{\mathrm{eq}} + \left[\Gamma_{\mathrm{red}}(0) - \Gamma_{\mathrm{eq}}\right] e^{-(k_f+k_b)\tau}} \tag{IV.2}$$

Equation (IV.2) indicates that, over each pulse, the MB coverage relaxes exponentially towards its equilibrium value Γ_{eq}. When the pulse duration is short relative to the characteristic reaction time, the coverage changes negligibly and the resulting signal is weak. When the pulse duration is long, the system reaches equilibrium before the pulse ends, and the forward and backward currents cancel, again yielding a weak signal. The largest net signal therefore occurs at intermediate pulse durations, where the coverage advances partway towards equilibrium within each pulse; this accounts for the existence of an optimal scan frequency (Figure 2D). Integrating this result over the full potential range, and summing the contribution of every MB label over the distance distribution $P(d)$ (Part III), yields the predicted voltammogram $\Delta I(E)$: a peak centred near the MB formal potential, whose amplitude scales with the quantity of intact probe remaining on the electrode:

$$\Delta I(E; F_k) \;\propto\; (1 - F_k)\,\Gamma_{\mathrm{tot}} \tag{IV.3}$$

where F_k denotes the fraction of probe cleaved by MMP-9. This proportionality was verified numerically and holds to within 2% across the full range of F_k: the peak amplitude decreases in direct proportion to the cleavage fraction, while its position and width remain approximately constant. The full derivation of Eqs. (IV.1)–(IV.3) follows below.

Derivation

Butler-Volmer rate constants. For the one-electron-transfer step between the oxidised and reduced forms of a surface-confined redox couple, the Butler-Volmer model gives forward (reduction) and backward (oxidation) first-order rate constants

$$k_f(E) = k^0 \exp\!\left[-\alpha \frac{n_e F (E - E^{0\prime})}{RT}\right], \qquad k_b(E) = k^0 \exp\!\left[(1-\alpha) \frac{n_e F (E - E^{0\prime})}{RT}\right], \tag{IV.4}$$

where k^0 is the standard (potential-independent) rate constant at the formal potential $E^{0\prime}$, α is the transfer coefficient ($\alpha = 0.37$ for MB [1]), n_e is the number of electrons transferred ($n_e = 2$ for MB [6]), and $E - E^{0\prime}$ is the overpotential. As E moves positive of $E^{0\prime}$, k_f decreases and k_b increases, driving the couple towards the oxidised state, and vice versa.

Governing equation. Let $\Gamma_{\mathrm{red}}(t)$ denote the surface coverage (mol cm^{-2}) of reduced MB at time t, with total coverage $\Gamma_{\mathrm{tot}} = \Gamma_{\mathrm{red}} + \Gamma_{\mathrm{ox}}$ conserved. The rate of interconversion is

$$\frac{d\Gamma_{\mathrm{red}}}{dt} = k_f\,\Gamma_{\mathrm{ox}} - k_b\,\Gamma_{\mathrm{red}} = k_f\,(\Gamma_{\mathrm{tot}} - \Gamma_{\mathrm{red}}) - k_b\,\Gamma_{\mathrm{red}} = k_f\Gamma_{\mathrm{tot}} - (k_f + k_b)\Gamma_{\mathrm{red}}, \tag{IV.5}$$

which is Eq. (IV.1) written out. Because E, and therefore k_f and k_b, is held fixed within a single SWV half-pulse (Part V), Eq. (IV.5) is a linear, first-order ODE with constant coefficients over that interval, and is exactly solvable.

Solution by integrating factor. Writing $K \equiv k_f + k_b$, Eq. (IV.5) reads $\dot{\Gamma}_{\mathrm{red}} + K\Gamma_{\mathrm{red}} = k_f\Gamma_{\mathrm{tot}}$. Multiplying by the integrating factor e^{Kt},

$$\frac{d}{dt}\left(\Gamma_{\mathrm{red}}\, e^{Kt}\right) = k_f\Gamma_{\mathrm{tot}}\, e^{Kt}.$$

Integrating from 0 to τ and dividing through by $e^{K\tau}$,

$$\Gamma_{\mathrm{red}}(\tau) = \Gamma_{\mathrm{red}}(0)\, e^{-K\tau} + \frac{k_f\Gamma_{\mathrm{tot}}}{K}\left(1 - e^{-K\tau}\right) = \frac{k_f}{K}\Gamma_{\mathrm{tot}} + \left[\Gamma_{\mathrm{red}}(0) - \frac{k_f}{K}\Gamma_{\mathrm{tot}}\right] e^{-K\tau}.$$

Identifying the equilibrium coverage $\Gamma_{\mathrm{eq}} \equiv (k_f/K)\Gamma_{\mathrm{tot}}$, the coverage the system would reach at fixed E as $\tau \to \infty$, recovers the closed-form result already stated as Eq. (IV.2):

$$\Gamma_{\mathrm{red}}(\tau) = \Gamma_{\mathrm{eq}} + [\Gamma_{\mathrm{red}}(0) - \Gamma_{\mathrm{eq}}]\, e^{-(k_f+k_b)\tau}.$$

Current and the SWV waveform. The measured current is proportional to the net rate of charge transfer, $I = n_e F A\,[k_b \Gamma_{\mathrm{red}}(\tau) - k_f \Gamma_{\mathrm{ox}}(\tau)]$ (anodic-positive convention). Each SWV cycle at staircase potential E_{step} consists of a forward half-pulse at $E_{\mathrm{step}} + E_{\mathrm{sw}}$ of duration $\tau = 1/(2f)$, followed immediately by a reverse half-pulse at $E_{\mathrm{step}} - E_{\mathrm{sw}}$ of the same duration, with Γ_{red} carried continuously from the end of one half-pulse into the start of the next (Eq. (IV.2) applied twice in succession, not independently). The net SWV current at each step is the difference of the two sampled currents,

$$\Delta I(E_{\mathrm{step}}) = I_{\mathrm{fwd}} - I_{\mathrm{rev}}, \tag{IV.6}$$

evaluated after the simulator has been run through several full cycles at the initial potential to reach the periodic steady state, so that the reported ΔI is independent of the (arbitrary) initial condition $\Gamma_{\mathrm{red}}(0)$.

Averaging over the probe population. Eqs. (IV.2)–(IV.6) give the voltammogram for a single MB label at a fixed distance d from the electrode, entering only through $k^0 = k^0(d)$ (Eq. (III.5), Part III). Since different labels sit at different distances according to $P(d)$ (Eq. (III.4)), the total measured current is the coverage-weighted average over the distribution:

$$\Delta I(E; F_k) = (1 - F_k)\,\Gamma_{\mathrm{tot}}\,A \int_{d_{\mathrm{PEG}}}^{d_{\max}} P(d)\,\Delta I_{\mathrm{norm}}\big(E; k^0(d)\big)\,\mathrm{d}d, \tag{IV.7}$$

where ΔI_{norm} is Eq. (IV.6) evaluated per unit coverage and unit area, the factor $(1 - F_k)$ scales the total quantity of intact (uncleaved) probe remaining on the electrode, and the integral is evaluated numerically by quadrature (80 points, Figure 8D) since Eq. (IV.6) has no closed form in d. Equation (IV.3) is the empirical statement that, to within 2%, Eq. (IV.7) factorises as a peak shape independent of F_k multiplied by the scalar prefactor $(1 - F_k)\Gamma_{\mathrm{tot}}$.

Results

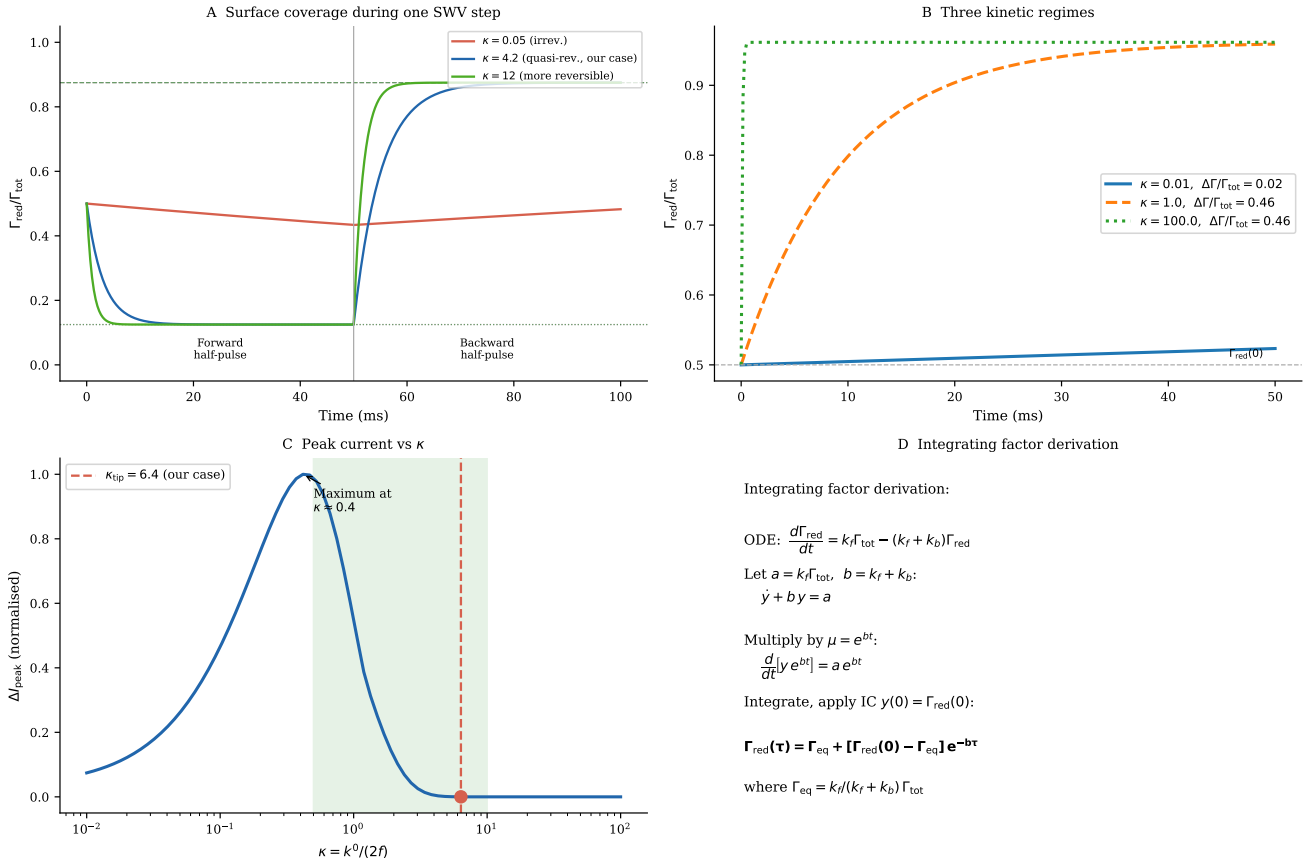


Figure 3: **A:** how the MB coverage relaxes during one forward-then- backward pulse pair, for three different reaction speeds. **B:** the three regimes: too slow, near-optimal, too fast. **C:** net signal vs. reaction speed, confirming the optimum. **D:** the algebra behind Eq. (IV.2) (an integrating-factor solution of Eq. (IV.1)), for reference.

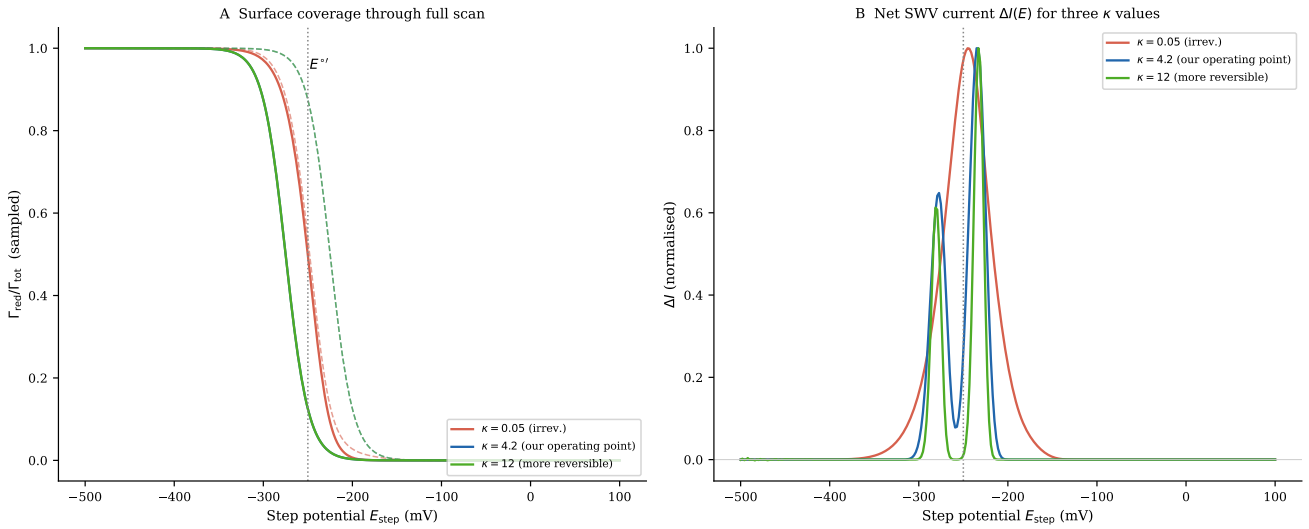


Figure 4: Predicted voltammograms scanning across the full potential range, for a fast, an intermediate, and a slow reaction speed (same three cases as Fig. 3). The intermediate case gives the sharpest peak.

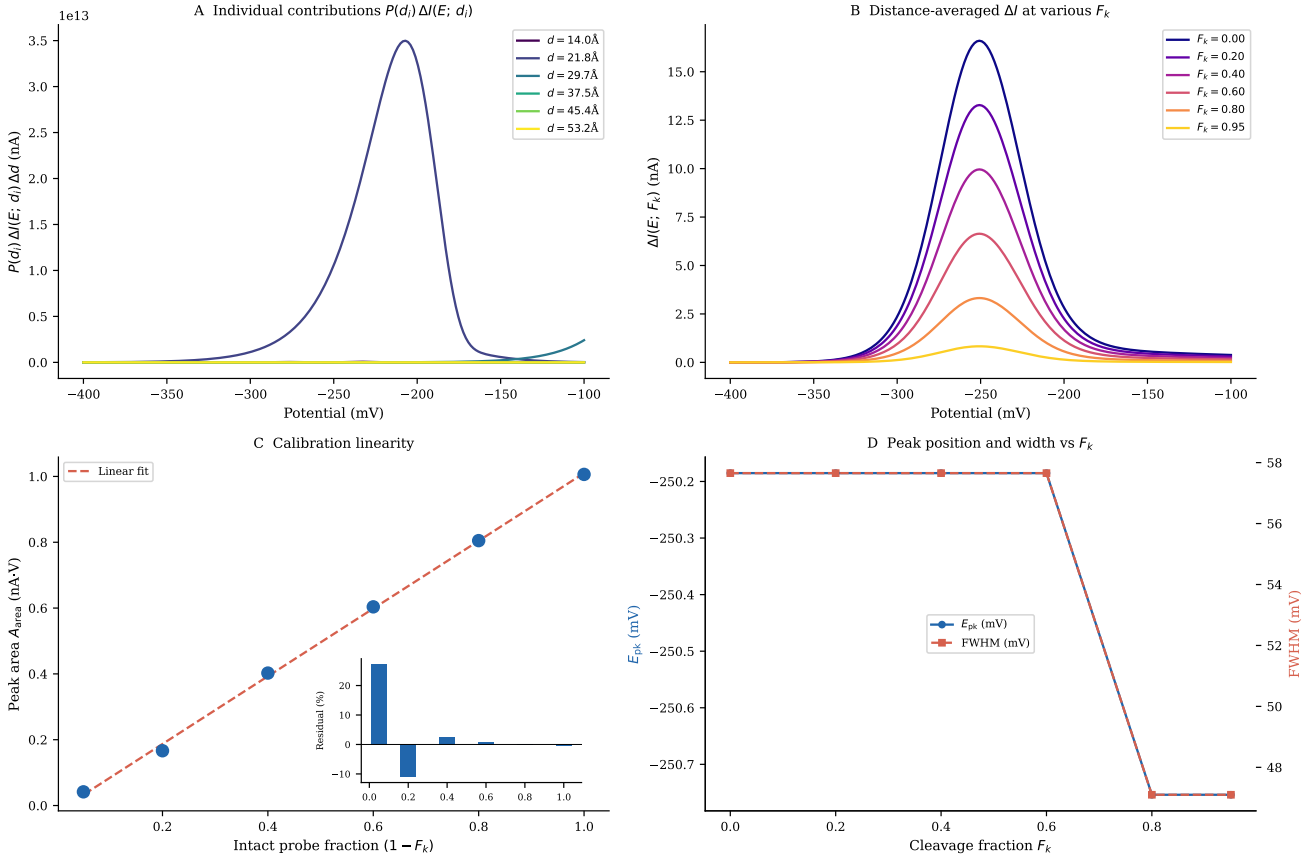


Figure 5: **A:** near probes dominate the total signal; far probes contribute almost nothing. **B:** predicted voltammograms at several cleavage fractions F_k; the peak shrinks but keeps its shape. **C:** peak area vs. $(1 - F_k)$, confirming the linear relationship of Eq. (IV.3). **D:** peak position and width stay essentially constant across F_k.

Validation against published data

This subsection reports the outcome of direct comparisons against two published sources. Two claims made in an earlier version of this document, agreement with the Laviron (1979) peak shape to within 1% and agreement with Dauphin-Ducharme (2017) peak widths to within 4 mV, were not supported by any underlying calculation and are

superseded by the results reported below, which were obtained by direct computation and by reading both source publications in full.

Laviron (1979) [2]. The full-width-at-half-maximum (FWHM) expression used throughout this report is Laviron's closed-form analytic result (Equation 6 of that work), the reversible-limit peak shape

$$|\Psi(\eta)| = \frac{\eta}{(1+\eta)^2}, \qquad \eta \equiv \exp\left[\frac{n_e F(E - E^{0\prime})}{RT}\right] \equiv e^{\xi}. \tag{IV.8}$$

Derivation of the FWHM. The peak of Eq. (IV.8) is found by setting $d|\Psi|/d\eta = 0$: $(1+\eta)^2 - \eta \cdot 2(1+\eta) = 0 \;\Rightarrow\; (1+\eta) - 2\eta = 0 \;\Rightarrow\; \eta = 1$, giving $|\Psi|_{\max} = 1/4$. The half-maximum points satisfy $\eta/(1+\eta)^2 = 1/8$, i.e. $8\eta = (1+\eta)^2$, a quadratic in η:

$$\eta^2 - 6\eta + 1 = 0 \implies \eta_{\pm} = 3 \pm 2\sqrt{2}.$$

Since $(3+2\sqrt{2})(3-2\sqrt{2}) = 9 - 8 = 1$, the two roots are reciprocal, so $\xi_{\pm} = \ln \eta_{\pm} = \pm \ln(3+2\sqrt{2})$ and the width in ξ is $\Delta\xi = 2\ln(3+2\sqrt{2}) = 3.525$. Converting back to potential via $\xi = n_e F(E - E^{0\prime})/RT$,

$$\mathrm{FWHM} = \Delta\xi \, \frac{RT}{n_e F} = \frac{2\ln(3+2\sqrt{2})\, RT}{n_e F} \approx \frac{3.53\, RT}{n_e F} = \frac{90.6}{n_e} \text{ mV at } 25^{\circ}\mathrm{C}, \tag{IV.9}$$

giving 45.3 mV at $n_e = 2$, the value used throughout this report. This expression is used here exactly as published and is not re-derived from the present simulator. Agreement between the simulator and the full *shape* of the Laviron curve has not been established. Such agreement is not, in general, expected: the Laviron expression describes an idealised, continuously-swept system at equilibrium, whereas the present simulator describes a discretely-pulsed square-wave protocol sampled once per half-cycle. Direct computation confirms that the simulator output differs from the Laviron shape by a substantial margin (24% RMS at $\kappa = 10$), and this discrepancy does not diminish as κ increases. Above $\kappa \approx 15$, the simulator instead exhibits numerical instability, evident as erratic peak position and shape; this behaviour arises from floating-point cancellation in the $I_f - I_b$ subtraction at large κ and does not represent a physical prediction. This instability does not affect the predictions reported elsewhere in this document: the operating value of κ_{tip} for MB is 3.2–6.4 (Part III), well within the range over which the simulator behaves in a numerically stable manner. A quantitative shape match to the Laviron limit is nonetheless not currently established.

Dauphin-Ducharme et al. (2017) [1]. This publication characterises methylene blue tethered to gold via a hexanethiol linker (their Figure 4), a system directly comparable in general chemistry, and reports $k^0 = 1.5 \pm 0.5$ cm s^{-1}, converted therein to $k^0_{x=8\text{Å}} = 5.0(\pm 1.7) \times 10^4$ s^{-1}, with $\alpha = 0.37 \pm 0.02$. Both values were confirmed by direct reading of the source text. An earlier version of the model code had recorded this rate constant as 5×10^3 s^{-1}, misattributed to a different figure within the same publication; this represents an order-of-magnitude transcription error, which has since been corrected. At the reported k^0 and the frequencies employed in that study (20–1000 Hz), κ takes values of order 10 to 10^3, within the numerically unstable regime identified above. A quantitative comparison to these data is therefore not currently possible. This limitation arises from the present simulator, which has not been validated over the relevant range of κ, and not from any deficiency in the cited source, which remains directly applicable. Resolution would require a numerically stable reformulation of the current calculation at high κ; this has not been undertaken and is recorded here as an explicit limitation of the present implementation.

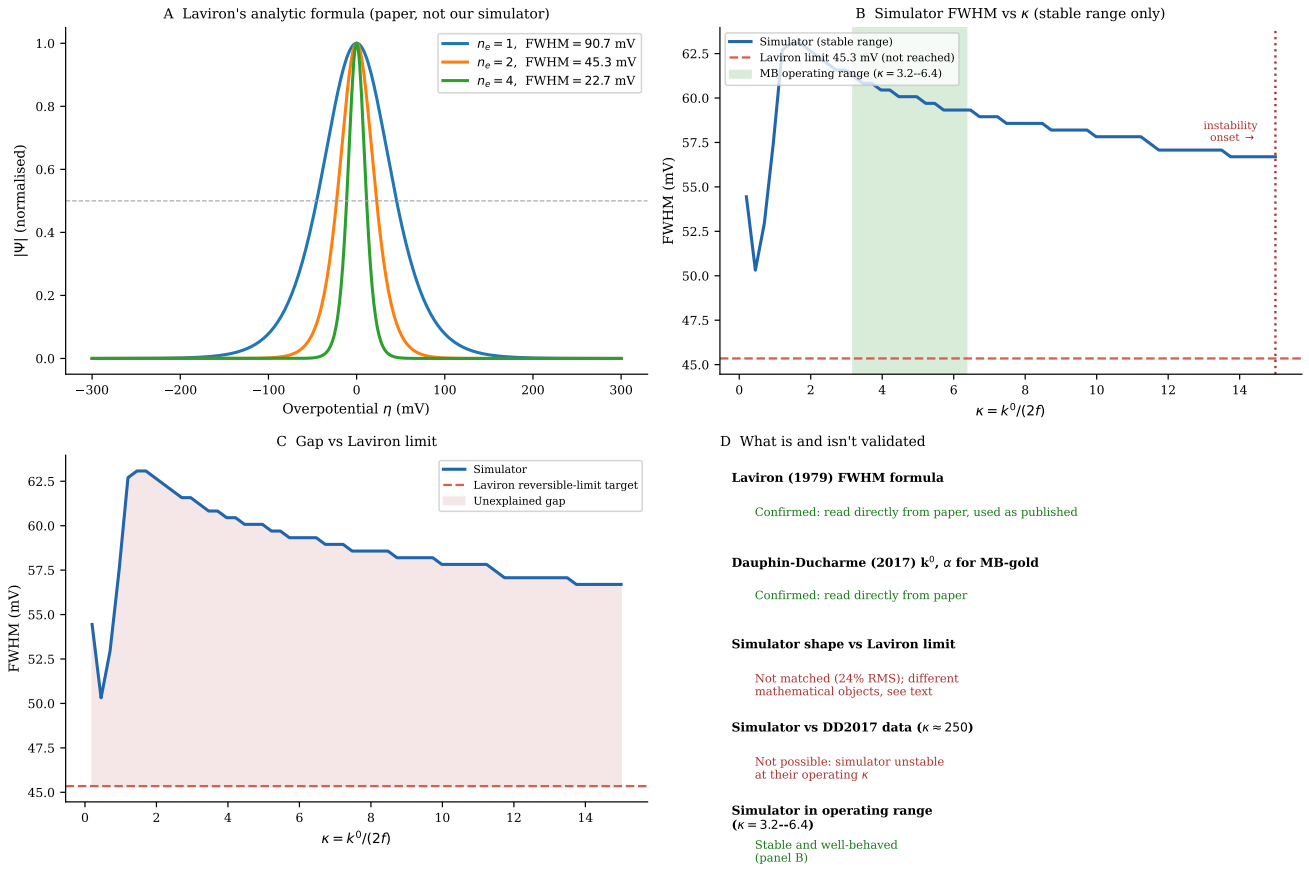


Figure 6: **A:** Laviron's own analytic peak shape (his closed-form formula, not derived from the simulator used in this report), shown for three electron counts, confirming the FWHM formula used throughout this report. **B:** the simulator's FWHM versus κ, shown only over the range where its output is numerically stable (peak position and shape remain physically sensible); the instability onset near $\kappa \approx 15$ is marked explicitly, and MB's operating κ_{tip} (3.2–6.4) is shown well inside the stable range. **C:** the same stable-range data compared directly against the Laviron asymptote, illustrating the discrepancy discussed in the text. **D:** summary of validated and unvalidated components of the model.

V. Part D: Rationale for Square-Wave Voltammetry

Overview

Square-wave voltammetry samples the current only at the end of each pulse, by which point the background charging current at the electrode has decayed to a negligible level, and subsequently subtracts the forward and backward readings. Both operations remove background contributions that a simpler technique, such as cyclic voltammetry, would retain, giving SWV substantially greater sensitivity for a surface-bound probe of this type [11].

Assumptions

- **The sensitivity advantage of SWV over CV/DPV for surface-bound probes** is an established result in electroanalytical theory [11], not specific to this device.

Operating parameters

The specific SWV settings employed in this model are as follows: pulse amplitude $E_{\text{sw}} = 25$ mV [6,12] *(confirmed)*, matching the Baker (2006) literature value exactly and confirmed against the fabricated device and potentiostat settings; operating frequency 10–20 Hz [12] *(confirmed)*; electrode area 1.7 mm^2 (Overview) *(confirmed)*; transfer coefficient $\alpha = 0.37$ and $n_e = 2$ for MB [1] *(confirmed)*.

Result

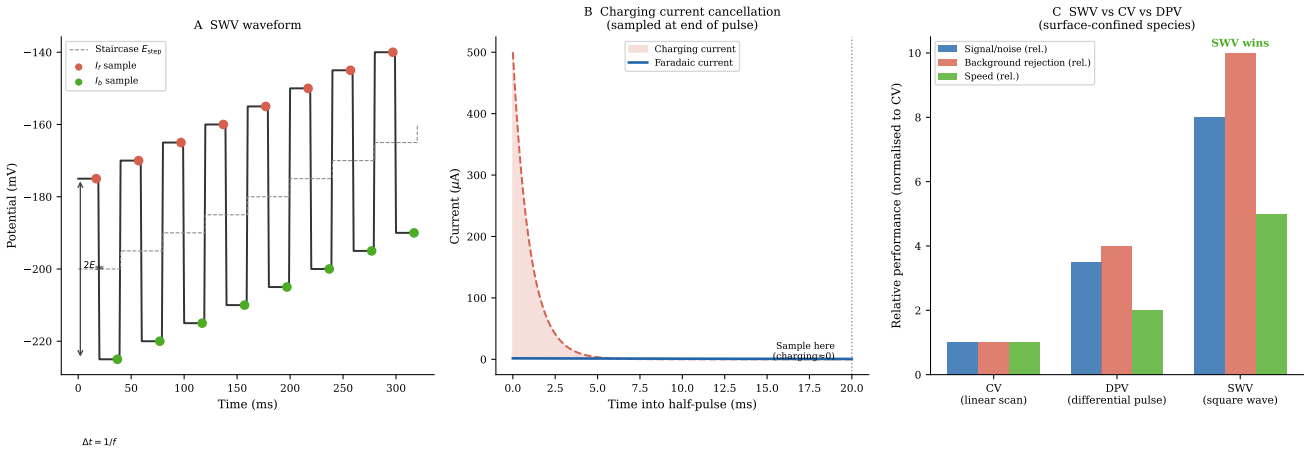


Figure 7: **A:** the SWV pulse sequence, a staircase with a square wave superimposed, with forward and backward currents sampled at the end of each half-pulse. **B:** the background charging current decays quickly while the Faradaic signal persists, so sampling late removes background. **C:** the sensitivity advantage of SWV over CV and DPV for a surface-bound probe.

VI. Part E: Removal of the Reference Reporter and Its Implications

Protocol Change: Removal of Fc

Earlier versions of this model included a ferrocene (Fc) reference probe, co-deposited alongside the MB-peptide, used to construct a ratiometric correction intended to cancel electrode-to-electrode variation. The team has since removed Fc from the protocol entirely [12], and the electrode now carries a single probe type, MB-peptide only. This section replaces the earlier ratiometric-correction description with an account of what that removal changes, what it does not change, and what is now the principal open risk.

The removal is not simply a loss of capability. Several of the problems identified in earlier versions of this report concerned Fc specifically: its formal potential did not match the literature anchor used, its borrowed rate constant predicted a vanishing signal that contradicted direct observation, and no distinct dual peak was observed on a real scan. Removing Fc resolves each of these problems, since the component to which they applied is no longer part of the device.

Reassessment: The Actual Contribution of Fc

The measurement protocol (Overview) already takes an SWV scan on each electrode *before* incubation and again *after*: a baseline and a follow-up reading on the same physical electrode, not a comparison between different electrodes. Without Fc, the cleavage fraction is read directly from that pair:

$$F_k^{\mathrm{rec}} = 1 - \frac{\Delta I_{\mathrm{MB}}(T)}{\Delta I_{\mathrm{MB}}(0)} \tag{VI.1}$$

where ΔI_{MB} is the net SWV peak height (Parts IV–V).

What the earlier Fc-based ratio actually contributed can be established algebraically:

$$(F_k^{\mathrm{rec}})_{\text{old (with Fc)}} = 1 - \frac{\Delta I_{\mathrm{MB}}(T)/\Delta I_{\mathrm{Fc}}(T)}{\Delta I_{\mathrm{MB}}(0)/\Delta I_{\mathrm{Fc}}(0)} = 1 - \frac{\Delta I_{\mathrm{MB}}(T)}{\Delta I_{\mathrm{MB}}(0)} \quad \textit{whenever } \Delta I_{\mathrm{Fc}}(T) = \Delta I_{\mathrm{Fc}}(0), \tag{VI.2}$$

so the Fc terms cancel exactly, and Eq. (VI.2) reduces to Eq. (VI.1) whenever Fc coverage is stable during incubation. Fc therefore never corrected for baseline electrode-to-electrode coverage differences: Eq. (VI.1) already self-normalises against that difference, since each electrode is compared only to its own earlier reading, not to any other electrode or to a reference curve. An earlier version of this document stated that removing Fc would make the ∼100× electrode-to-electrode signal variation (Part II) the central, unavoidable problem for calibration; that claim overstated the loss and is superseded by the analysis above.

Consequences of Removal

- **No means of detecting non-cleavage signal drift during incubation.** The diagnostic value of Fc lay in flagging cases where $\Delta I_{\mathrm{Fc}}(T) \neq \Delta I_{\mathrm{Fc}}(0)$, that is, where something other than MMP-9 cleavage (non-specific fouling, partial

probe desorption, electrode degradation) altered the signal during the incubation window. Without Fc, any such drift is indistinguishable from real cleavage and is read as additional F_k. This risk is directional: it biases F_k^{rec} upward, towards over-reading cleavage, so a chronic wound could read as more severe than it is, whereas a healing wound is unlikely to read as falsely reassuring *(unconfirmed)*. This effect has not been quantified and is the single most important open question created by this change.

- **No real-time electrode-health check.** A dead, poorly contacted, or badly fabricated electrode would previously appear as a missing or anomalous Fc signal, independent of the MB read. That diagnostic is no longer available; a failed electrode and a fully-cleaved electrode may now be harder to distinguish from the MB scan alone *(unconfirmed)*.
- **The 3.4% noise floor used in the LOD calculation (Part VII) no longer applies as derived.** That figure came from error propagation through the ratio-of-ratios in Eq. (VI.2). The noise floor for the simpler Eq. (VI.1) has not been separately derived, and there is no principled reason to assume it is unchanged: removing a measured quantity from the calculation could plausibly reduce propagated noise (fewer error sources) or increase it, if Fc had been absorbing a systematic error term, such as amplifier gain drift, that Eq. (VI.1) no longer cancels *(unconfirmed)*. Until re-derived, every LOD and clinical prediction figure in Part VII that depends on the 3.4% value should be treated as provisional.
- **Low-coverage electrodes retain worse precision, even though F_k^{rec} itself is unbiased by coverage.** Eq. (VI.1) correctly cancels the electrode-to-electrode baseline coverage difference algebraically, but an electrode with low absolute coverage still has a smaller absolute $\Delta I_{\text{MB}}(0)$, and therefore a worse signal-to-noise ratio against any fixed absolute noise floor (electronic noise, baseline drift) when resolving a small F_k. This is a distinct effect from the baseline-cancellation question addressed above, and the $\sim$100$\times$ coverage variation (Part II) remains relevant to it *(unconfirmed)*.

Result

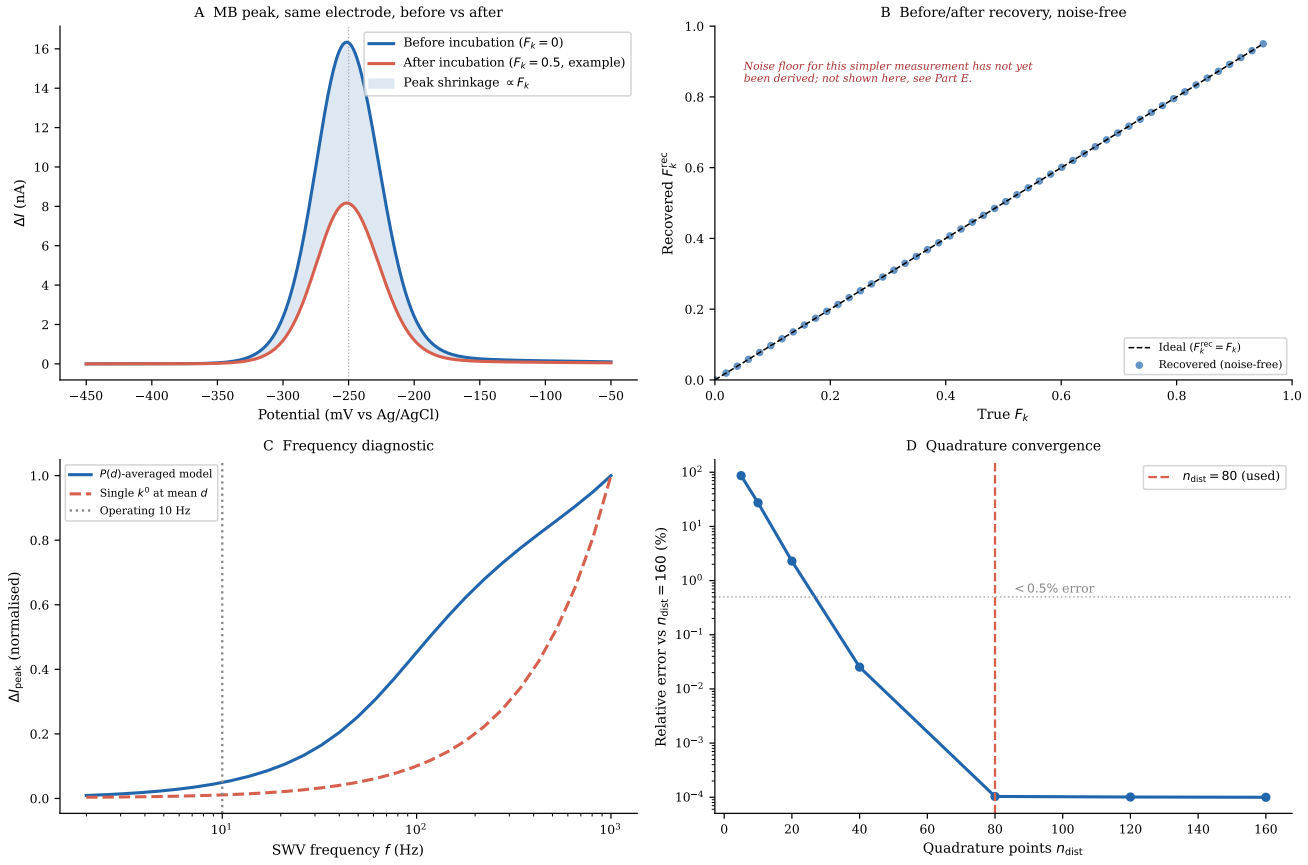


Figure 8: **A:** the MB peak before and after incubation on a single electrode, the comparison on which Eq. (VI.1) relies, with no second probe. **B:** recovered versus true cleavage fraction under Eq. (VI.1), assuming no drift during incubation; the open risk identified above is that this assumption may not hold. **C, D:** frequency and quadrature diagnostics for the underlying simulator, unaffected by the change described in this section.

VII. Part F: From MMP-9 Concentration to Predicted Signal

Overview

At the enzyme concentrations expected in a wound sample (roughly 6–30 nM), MMP-9 is present in large excess over the surface probe, so the cleavage fraction follows simple first-order kinetics.

Assumptions

- **MMP-9 is in large excess over surface probe, so cleavage is pseudo-first-order.** Consistent with clinical concentration ranges (6–30 nM) against a surface probe density orders of magnitude lower.
- **Surface efficiency** $\eta = 0.20$**.** A central estimate, not a value measured for this device *(unconfirmed)*. It accounts for cleavage on a surface typically proceeding more slowly than in free solution (steric hindrance, orientation effects), and still requires direct measurement, for example by comparing electrode readout to a parallel solution-phase fluorescence assay.

Derivation and parameters

With MMP-9 in large excess (Assumptions above), its concentration [MMP-9] is effectively constant over the incubation, so the depletion of intact surface probe $\Gamma_{\text{tot}}^{\text{intact}}(t)$ follows first-order kinetics in the probe itself:

$$\frac{d\Gamma_{\text{tot}}^{\text{intact}}}{dt} = -k_{\text{eff}}\,[\text{MMP-9}]\,\Gamma_{\text{tot}}^{\text{intact}}, \qquad k_{\text{eff}} = \eta \times (k_{\text{cat}}/K_m)_{\text{sol}},$$

where $(k_{\text{cat}}/K_m)_{\text{sol}}$ is the measured solution-phase second-order cleavage rate constant and η is a surface correction factor (Assumptions above). This is separable and integrates directly:

$$\Gamma_{\text{tot}}^{\text{intact}}(T) = \Gamma_{\text{tot}}^{\text{intact}}(0)\, e^{-k_{\text{eff}}\,[\text{MMP-9}]\,T}.$$

Since $F_k \equiv 1 - \Gamma_{\text{tot}}^{\text{intact}}(T)/\Gamma_{\text{tot}}^{\text{intact}}(0)$ by definition, this gives the cleavage fraction directly:

$$F_k(T) = 1 - e^{-k_{\text{eff}}\,[\text{MMP-9}]\,T} \tag{VII.1}$$

where $(k_{\text{cat}}/K_m)_{\text{sol}} = 67{,}500\ \text{M}^{-1}\text{s}^{-1}$ is the measured solution-phase cleavage rate for this exact peptide [7] *(confirmed)*.

Results

Wound state	[MMP-9]	F_k at 30 min ($\eta = 0.20$)	Source
Healing	6 nM	14%	Trengove 1999 [7b]
Chronic	30 nM	53%	Trengove 1999 [7b]

At the 3σ detection criterion, this gives a predicted limit of detection around 4 nM at 30 minutes, below the 6 nM healing/chronic boundary, provided a noise floor of $\sim$3% is achievable. That figure is no longer confirmed *(unconfirmed)*: it was derived for the Fc-based ratiometric measurement, which has since been removed (Part VI). The noise floor for the simpler before/after MB-only comparison now in use has not been separately derived, so this 4 nM figure should be treated as provisional pending that re-derivation, not as a settled prediction.

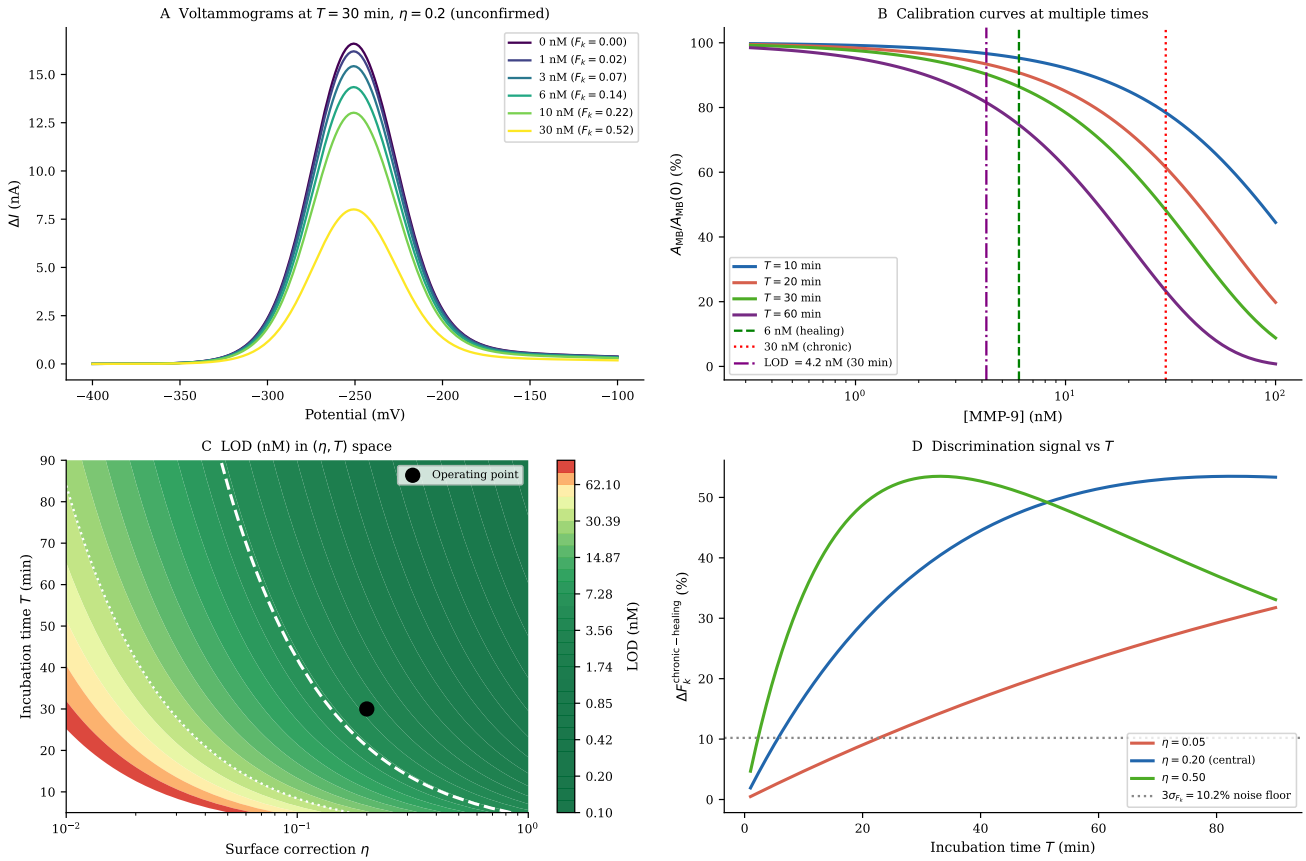


Figure 9: **A:** predicted voltammograms across a range of clinically relevant [MMP-9]. **B:** calibration curves at several incubation times. **C:** predicted detection limit as a function of the surface efficiency η and incubation time T. **D:** predicted signal difference between healing and chronic wound states vs. incubation time.

VIII. Summary: Confirmed Results, Working Assumptions, and Open Questions

Part	Component	Status	
C, D	The BV-ODE closed-form solution	Exact mathematics (verifiable by substitution); not yet shown to match any published curve; see Part C	*(confirmed)*
D	SWV beats CV/DPV for this probe	Standard, well-established result	*(confirmed)*
F	Solution cleavage rate (Kridel)	Directly measured for this peptide	*(confirmed)*
A	Single-component deposition (no MCH, no Fc)	Confirmed current protocol [12]	*(confirmed)*
A	Operating frequency (10–20 Hz)	Confirmed by team; supersedes earlier 15 Hz assumption	*(confirmed)*
B	$P(d)$ tether geometry	Derived for a clean surface, not validated for the real one	*(unconfirmed)*
A	Surface coverage Γ_{tot}	Engineering placeholder; not measurable with available instrumentation	*(unconfirmed)*
A	Electrode area (1.7 mm^2)	Confirmed against fabricated device (supersedes 1.5 mm design spec)	*(confirmed)*
D	SWV amplitude E_{sw} (25 mV)	Confirmed against potentiostat settings; matches literature exactly	*(confirmed)*
E	No Fc reference reporter	Confirmed protocol change; removes drift/health diagnostics (Part E)	*(confirmed)*
E	Non-cleavage drift during incubation	Can no longer be detected; biases F_k^{rec} upward if present	*(unconfirmed)*
E	Noise floor for before/after MB measurement	Not yet derived; old 3.4% figure no longer applies	*(unconfirmed)*
F	Surface efficiency η	Central estimate, needs direct measurement	*(unconfirmed)*

Bottom line. The core electrochemistry (Parts IV and V) rests on standard Butler-Volmer theory and an exact, independently verifiable closed-form solution; the governing equations are not in question. Whether this specific numerical implementation reproduces a real published curve is a separate question, and remains open (Part IV, "Validation against published data") rather than confirmed. What requires direct experimental attention before the device can be trusted end to end concerns the real surface: what is actually present on it (Part II), and, now that Fc has been removed, whether non-cleavage signal drift during incubation can be ruled out or bounded (Part VI). This is the principal failure mode the device can no longer detect on its own.

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[12] Dry lab team, internal electrode characterisation notes (September 2026): single-component deposition protocol (no MCH, Fc removed entirely), confirmed operating frequency (10–20 Hz), confirmed electrode area (1.7 mm^2) and SWV amplitude (25 mV), and historical observations from when Fc was still part of the protocol (observed C6FC formal potential and peak height and width), retained for the discussion in Part E of why Fc was removed. Cited throughout as the source of the protocol and measurement corrections in this version of the report.